

Lecture: Integration of Rational Functions (Rational Fractions)

Goal: Learn how to integrate any rational function $\frac{P(x)}{Q(x)}$ using a systematic method.

Introduction

A **rational function** is a ratio of two polynomials:

$$R(x) = \frac{P(x)}{Q(x)}.$$

To integrate rational functions, we follow this universal strategy:

- Make the rational function proper.
- Factor the denominator.
- Apply the correct partial fraction form.
- Integrate each term using standard formulas.

Proper vs Improper Rational Function

Improper Rational Function

$$\deg P(x) \geq \deg Q(x)$$

Perform **polynomial long division** before integrating.

Proper Rational Function

$$\deg P(x) < \deg Q(x)$$

Proceed directly to partial fractions.

Factor the Denominator

Typical factor types:

- Linear: $(ax + b)$
- Repeated linear: $(ax + b)^n$
- Irreducible quadratic: $ax^2 + px + q$
- Repeated quadratic: $(ax^2 + px + q)^n$

Partial Fraction Forms

(a) **Linear factor** $(ax + b)$

$$\frac{A}{ax + b}$$

(b) **Repeated linear factor** $(ax + b)^n$

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_n}{(ax + b)^n}$$

(c) **Irreducible quadratic factor** $(ax^2 + px + q)$

$$\frac{Ax + B}{ax^2 + px + q}$$

(d) **Repeated quadratic factor**

$$\frac{A_1x + B_1}{ax^2 + px + q} + \frac{A_2x + B_2}{(ax^2 + px + q)^2} + \cdots$$

Examples Covering All Rules

A. Linear Factors

Example 1.

$$\int \frac{3}{x-5} dx$$
$$\int \frac{3}{x-5} dx = 3 \int \frac{1}{x-5} dx = 3 \ln |x-5| + C.$$

$$\boxed{\int \frac{3}{x-5} dx = 3 \ln |x-5| + C}$$

Example 2.

$$\int \frac{5x-1}{(x-2)(x+3)} dx$$

First use partial fractions. We assume

$$\frac{5x-1}{(x-2)(x+3)} = \frac{A}{x-2} + \frac{B}{x+3}.$$

Then

$$5x-1 = A(x+3) + B(x-2) = (A+B)x + (3A-2B).$$

Equating coefficients:

$$A+B=5, \quad 3A-2B=-1.$$

Solve:

$$A=5-B, \quad 3(5-B)-2B=-1 \Rightarrow 15-3B-2B=-1 \Rightarrow 15-5B=-1 \Rightarrow 5B=16 \Rightarrow B=\frac{16}{5}.$$

Then

$$A=5-\frac{16}{5}=\frac{25}{5}-\frac{16}{5}=\frac{9}{5}.$$

So

$$\frac{5x-1}{(x-2)(x+3)} = \frac{\frac{9}{5}}{x-2} + \frac{\frac{16}{5}}{x+3}.$$

Thus

$$\begin{aligned} \int \frac{5x-1}{(x-2)(x+3)} dx &= \frac{9}{5} \int \frac{dx}{x-2} + \frac{16}{5} \int \frac{dx}{x+3} \\ &= \frac{9}{5} \ln |x-2| + \frac{16}{5} \ln |x+3| + C. \end{aligned}$$

$$\boxed{\int \frac{5x-1}{(x-2)(x+3)} dx = \frac{9}{5} \ln |x-2| + \frac{16}{5} \ln |x+3| + C}$$

 Example 3.

$$\int \frac{\cos x}{(a + \sin x)(b - \sin x)} dx$$

Let

$$t = \sin x \quad \Rightarrow \quad dt = \cos x \, dx.$$

Then

$$\int \frac{\cos x}{(a + \sin x)(b - \sin x)} dx = \int \frac{dt}{(a + t)(b - t)}.$$

Decompose:

$$\frac{1}{(a + t)(b - t)} = \frac{A}{a + t} + \frac{B}{b - t}.$$

Then

$$1 = A(b - t) + B(a + t) = (Ab + Ba) + (B - A)t.$$

So

$$B - A = 0 \Rightarrow B = A, \quad A(b + a) = 1 \Rightarrow A = \frac{1}{a + b}.$$

Hence

$$\frac{1}{(a + t)(b - t)} = \frac{1}{a + b} \left(\frac{1}{a + t} + \frac{1}{b - t} \right).$$

Thus

$$\begin{aligned} \int \frac{dt}{(a + t)(b - t)} &= \frac{1}{a + b} \int \left(\frac{1}{a + t} + \frac{1}{b - t} \right) dt \\ &= \frac{1}{a + b} (\ln |a + t| - \ln |b - t|) + C. \end{aligned}$$

Substitute $t = \sin x$:

$$\boxed{\int \frac{\cos x}{(a + \sin x)(b - \sin x)} dx = \frac{1}{a + b} \ln \left| \frac{a + \sin x}{b - \sin x} \right| + C}$$

B. Repeated Linear Factors

Example 4.

$$\int \frac{1}{(x+1)^2} dx$$

We use

$$\int (x+1)^{-2} dx = -(x+1)^{-1} + C.$$

So

$$\boxed{\int \frac{1}{(x+1)^2} dx = -\frac{1}{x+1} + C}$$

Example 5.

$$\int \frac{5x+4}{(x-1)^2} dx$$

Write

$$\frac{5x+4}{(x-1)^2} = \frac{5(x-1)+9}{(x-1)^2} = \frac{5}{x-1} + \frac{9}{(x-1)^2}.$$

Then

$$\begin{aligned} \int \frac{5x+4}{(x-1)^2} dx &= 5 \int \frac{dx}{x-1} + 9 \int \frac{dx}{(x-1)^2} \\ &= 5 \ln|x-1| - \frac{9}{x-1} + C. \end{aligned}$$

$$\boxed{\int \frac{5x+4}{(x-1)^2} dx = 5 \ln|x-1| - \frac{9}{x-1} + C}$$

Example 6.

$$\int \frac{x^2}{(x+1)^2(x+2)} dx$$

Decompose:

$$\frac{x^2}{(x+1)^2(x+2)} = \frac{A}{x+2} + \frac{B}{x+1} + \frac{C}{(x+1)^2}.$$

Then

$$x^2 = A(x+1)^2 + B(x+1)(x+2) + C(x+2).$$

Expand:

$$(x+1)^2 = x^2 + 2x + 1, \quad (x+1)(x+2) = x^2 + 3x + 2.$$

So

$$x^2 = A(x^2 + 2x + 1) + B(x^2 + 3x + 2) + C(x + 2).$$

$$x^2 = (A+B)x^2 + (2A+3B+C)x + (A+2B+2C).$$

Equate coefficients:

$$\begin{cases} A+B=1, \\ 2A+3B+C=0, \\ A+2B+2C=0. \end{cases}$$

Solving gives

$$A=4, \quad B=-3, \quad C=1.$$

Thus

$$\frac{x^2}{(x+1)^2(x+2)} = \frac{4}{x+2} - \frac{3}{x+1} + \frac{1}{(x+1)^2}.$$

Hence

$$\begin{aligned} \int \frac{x^2}{(x+1)^2(x+2)} dx &= 4 \int \frac{dx}{x+2} - 3 \int \frac{dx}{x+1} + \int \frac{dx}{(x+1)^2} \\ &= 4 \ln|x+2| - 3 \ln|x+1| - \frac{1}{x+1} + C. \end{aligned}$$

$$\boxed{\int \frac{x^2}{(x+1)^2(x+2)} dx = 4 \ln|x+2| - 3 \ln|x+1| - \frac{1}{x+1} + C}$$

C. Irreducible Quadratic Factors

Example 7.

$$\int \frac{1}{x^2 + 4} dx$$

We use the standard form $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$ with $a = 2$:

$$\boxed{\int \frac{1}{x^2 + 4} dx = \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C}$$

Example 8.

$$\int \frac{3x + 1}{x^2 + 2x + 5} dx$$

Complete the square:

$$x^2 + 2x + 5 = (x + 1)^2 + 4.$$

Rewrite the numerator:

$$3x + 1 = \frac{3}{2}(2x + 2) - 2.$$

So

$$\int \frac{3x + 1}{x^2 + 2x + 5} dx = \frac{3}{2} \int \frac{2x + 2}{x^2 + 2x + 5} dx - 2 \int \frac{dx}{x^2 + 2x + 5}.$$

For the first integral, let $u = x^2 + 2x + 5$, then $du = (2x + 2) dx$:

$$\frac{3}{2} \int \frac{2x + 2}{x^2 + 2x + 5} dx = \frac{3}{2} \int \frac{du}{u} = \frac{3}{2} \ln|u| + C = \frac{3}{2} \ln(x^2 + 2x + 5).$$

For the second integral, use the standard form with $(x + 1)^2 + 2^2$:

$$\int \frac{dx}{x^2 + 2x + 5} = \int \frac{dx}{(x + 1)^2 + 4} = \frac{1}{2} \arctan\left(\frac{x + 1}{2}\right) + C.$$

Hence

$$-2 \int \frac{dx}{x^2 + 2x + 5} = -2 \cdot \frac{1}{2} \arctan\left(\frac{x + 1}{2}\right) = -\arctan\left(\frac{x + 1}{2}\right).$$

Therefore

$$\boxed{\int \frac{3x + 1}{x^2 + 2x + 5} dx = \frac{3}{2} \ln(x^2 + 2x + 5) - \arctan\left(\frac{x + 1}{2}\right) + C}$$

D. Repeated Quadratic Factors

Example 9.

$$\int \frac{x}{(x^2 + 1)^2} dx$$

Let

$$u = x^2 + 1 \quad \Rightarrow \quad du = 2x dx \quad \Rightarrow \quad x dx = \frac{1}{2} du.$$

Then

$$\int \frac{x}{(x^2 + 1)^2} dx = \int \frac{1}{u^2} \cdot \frac{1}{2} du = \frac{1}{2} \int u^{-2} du = \frac{1}{2} (-u^{-1}) + C = -\frac{1}{2u} + C.$$

Substitute $u = x^2 + 1$:

$$\boxed{\int \frac{x}{(x^2 + 1)^2} dx = -\frac{1}{2(x^2 + 1)} + C}$$

Example 10.

$$\int \frac{3x^2 + 1}{(x^2 + 1)^2} dx$$

Rewrite the numerator:

$$3x^2 + 1 = 3(x^2 + 1) - 2.$$

Thus

$$\int \frac{3x^2 + 1}{(x^2 + 1)^2} dx = 3 \int \frac{1}{x^2 + 1} dx - 2 \int \frac{1}{(x^2 + 1)^2} dx.$$

We use:

$$\int \frac{dx}{x^2 + 1} = \arctan x + C,$$

and the standard exact result

$$\int \frac{dx}{(x^2 + 1)^2} = \frac{x}{2(x^2 + 1)} + \frac{1}{2} \arctan x + C.$$

Hence

$$3 \int \frac{1}{x^2 + 1} dx = 3 \arctan x,$$

and

$$-2 \int \frac{dx}{(x^2 + 1)^2} = -2 \left(\frac{x}{2(x^2 + 1)} + \frac{1}{2} \arctan x \right) = -\frac{x}{x^2 + 1} - \arctan x.$$

Combine:

$$\int \frac{3x^2 + 1}{(x^2 + 1)^2} dx = 3 \arctan x - \frac{x}{x^2 + 1} - \arctan x = 2 \arctan x - \frac{x}{x^2 + 1} + C.$$

$$\boxed{\int \frac{3x^2 + 1}{(x^2 + 1)^2} dx = 2 \arctan x - \frac{x}{x^2 + 1} + C}$$

 Example 11.

$$\int \frac{5x + 7}{(x^2 + 4)^2} dx$$

Rewrite numerator to match derivatives:

$$5x + 7 = \frac{5}{2}(2x) + 7.$$

So

$$\frac{5x + 7}{(x^2 + 4)^2} = \frac{5}{2} \cdot \frac{2x}{(x^2 + 4)^2} + \frac{7}{(x^2 + 4)^2}.$$

First part: Let $u = x^2 + 4$, then $du = 2x dx$.

$$\frac{5}{2} \int \frac{2x}{(x^2 + 4)^2} dx = \frac{5}{2} \int \frac{du}{u^2} = \frac{5}{2}(-u^{-1}) + C = -\frac{5}{2(x^2 + 4)}.$$

Second part: use the exact formula

$$\int \frac{dx}{(x^2 + a^2)^2} = \frac{x}{2a^2(x^2 + a^2)} + \frac{1}{2a^3} \arctan\left(\frac{x}{a}\right) + C.$$

Here $a^2 = 4$, so $a = 2$:

$$\int \frac{dx}{(x^2 + 4)^2} = \frac{x}{8(x^2 + 4)} + \frac{1}{16} \arctan\left(\frac{x}{2}\right).$$

Therefore

$$\int \frac{7}{(x^2 + 4)^2} dx = 7 \int \frac{dx}{(x^2 + 4)^2} = \frac{7x}{8(x^2 + 4)} + \frac{7}{16} \arctan\left(\frac{x}{2}\right).$$

Combine:

$$\int \frac{5x + 7}{(x^2 + 4)^2} dx = -\frac{5}{2(x^2 + 4)} + \frac{7x}{8(x^2 + 4)} + \frac{7}{16} \arctan\left(\frac{x}{2}\right) + C.$$

Put over common denominator $8(x^2 + 4)$:

$$-\frac{5}{2(x^2 + 4)} = -\frac{20}{8(x^2 + 4)}.$$

So the rational part is

$$\frac{7x - 20}{8(x^2 + 4)}.$$

Thus

$$\boxed{\int \frac{5x + 7}{(x^2 + 4)^2} dx = \frac{7x - 20}{8(x^2 + 4)} + \frac{7}{16} \arctan\left(\frac{x}{2}\right) + C}$$

E. Mixed Linear + Quadratic

Example 12.

$$\int \frac{2x+5}{x(x^2+4)} dx$$

Partial fractions:

$$\frac{2x+5}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}.$$

Then

$$2x+5 = A(x^2+4) + x(Bx+C) = (A+B)x^2 + Cx + 4A.$$

Equate:

$$\begin{cases} A+B=0, \\ C=2, \\ 4A=5. \end{cases}$$

So

$$A = \frac{5}{4}, \quad B = -\frac{5}{4}, \quad C = 2.$$

Thus

$$\frac{2x+5}{x(x^2+4)} = \frac{5/4}{x} - \frac{5x/4}{x^2+4} + \frac{2}{x^2+4}.$$

Then

$$\int \frac{2x+5}{x(x^2+4)} dx = \frac{5}{4} \int \frac{dx}{x} - \frac{5}{4} \int \frac{x}{x^2+4} \cdot 2 \frac{dx}{2} + 2 \int \frac{dx}{x^2+4}.$$

More directly:

$$\begin{aligned} \int \frac{5}{4} \frac{dx}{x} &= \frac{5}{4} \ln|x|, \\ \int \frac{5x}{4(x^2+4)} dx &= \frac{5}{8} \int \frac{2x}{x^2+4} dx = \frac{5}{8} \ln(x^2+4), \\ \int \frac{2}{x^2+4} dx &= 2 \cdot \frac{1}{2} \arctan\left(\frac{x}{2}\right) = \arctan\left(\frac{x}{2}\right). \end{aligned}$$

So

$$\int \frac{2x+5}{x(x^2+4)} dx = \frac{5}{4} \ln|x| - \frac{5}{8} \ln(x^2+4) + \arctan\left(\frac{x}{2}\right) + C.$$

$$\int \frac{2x+5}{x(x^2+4)} dx = \frac{5}{4} \ln|x| - \frac{5}{8} \ln(x^2+4) + \arctan\left(\frac{x}{2}\right) + C$$

F. Improper Rational Function

Example 13.

$$\int \frac{x^3 + 2}{(x - 1)^2} dx$$

This is improper: $\deg(\text{numerator}) = 3$, $\deg(\text{denominator}) = 2$.

Perform polynomial long division of $x^3 + 2$ by $(x - 1)^2 = x^2 - 2x + 1$.

$$\frac{x^3 + 2}{x^2 - 2x + 1} = x + 2 + \frac{3x}{x^2 - 2x + 1}.$$

Thus

$$\frac{x^3 + 2}{(x - 1)^2} = x + 2 + \frac{3x}{(x - 1)^2}.$$

Now decompose

$$\frac{3x}{(x - 1)^2} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2}.$$

Then

$$3x = A(x - 1) + B = Ax - A + B.$$

Equate:

$$A = 3, \quad -A + B = 0 \Rightarrow B = 3.$$

So

$$\frac{3x}{(x - 1)^2} = \frac{3}{x - 1} + \frac{3}{(x - 1)^2}.$$

Therefore

$$\frac{x^3 + 2}{(x - 1)^2} = x + 2 + \frac{3}{x - 1} + \frac{3}{(x - 1)^2}.$$

Integrate term-by-term:

$$\begin{aligned} \int \frac{x^3 + 2}{(x - 1)^2} dx &= \int x dx + \int 2 dx + 3 \int \frac{dx}{x - 1} + 3 \int \frac{dx}{(x - 1)^2} \\ &= \frac{x^2}{2} + 2x + 3 \ln |x - 1| - \frac{3}{x - 1} + C. \end{aligned}$$

$$\boxed{\int \frac{x^3 + 2}{(x - 1)^2} dx = \frac{x^2}{2} + 2x + 3 \ln |x - 1| - \frac{3}{x - 1} + C}$$

Mixed Case

Example 14.

$$\int \frac{x^4 + 3x^2 + 5}{x(x^2 + 1)^2} dx$$

We look for

$$\frac{x^4 + 3x^2 + 5}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}.$$

Multiply both sides by $x(x^2 + 1)^2$:

$$x^4 + 3x^2 + 5 = A(x^2 + 1)^2 + x(Bx + C)(x^2 + 1) + x(Dx + E).$$

Compute:

$$(x^2 + 1)^2 = x^4 + 2x^2 + 1,$$

$$x(Bx + C)(x^2 + 1) = x(Bx^3 + Bx + Cx^2 + C) = Bx^4 + Cx^3 + Bx^2 + Cx,$$

$$x(Dx + E) = Dx^2 + Ex.$$

So

$$x^4 + 3x^2 + 5 = A(x^4 + 2x^2 + 1) + (Bx^4 + Cx^3 + Bx^2 + Cx) + (Dx^2 + Ex).$$

Collect:

$$x^4 + 3x^2 + 5 = (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A.$$

Match coefficients with $x^4 + 0x^3 + 3x^2 + 0x + 5$:

$$\begin{cases} A + B = 1, \\ C = 0, \\ 2A + B + D = 3, \\ C + E = 0 \Rightarrow E = -C = 0, \\ A = 5. \end{cases}$$

From $A = 5$ and $A + B = 1$:

$$B = 1 - 5 = -4.$$

Then

$$2A + B + D = 3 \Rightarrow 10 - 4 + D = 3 \Rightarrow D = -3.$$

And $C = 0, E = 0$.

Hence

$$\frac{x^4 + 3x^2 + 5}{x(x^2 + 1)^2} = \frac{5}{x} - \frac{4x}{x^2 + 1} - \frac{3x}{(x^2 + 1)^2}.$$

Integrate each term:

$$\int \frac{5}{x} dx = 5 \ln |x|,$$

$$\int \frac{-4x}{x^2+1} dx = -4 \cdot \frac{1}{2} \int \frac{2x}{x^2+1} dx = -2 \ln(x^2+1),$$

$$\int \frac{-3x}{(x^2+1)^2} dx.$$

For the last one, set $u = x^2 + 1$, $du = 2x dx$, so $x dx = \frac{1}{2}du$:

$$\int \frac{-3x}{(x^2+1)^2} dx = -3 \int \frac{x}{(x^2+1)^2} dx = -3 \cdot \frac{1}{2} \int u^{-2} du = -\frac{3}{2} (-u^{-1}) + C = \frac{3}{2(x^2+1)}.$$

Thus

$$\int \frac{x^4 + 3x^2 + 5}{x(x^2+1)^2} dx = 5 \ln|x| - 2 \ln(x^2+1) + \frac{3}{2(x^2+1)} + C.$$

$$\boxed{\int \frac{x^4 + 3x^2 + 5}{x(x^2+1)^2} dx = 5 \ln|x| - 2 \ln(x^2+1) + \frac{3}{2(x^2+1)} + C}$$